Exercise 10.1

An N channel MOSFET

t_{ox} = 3 nm
L = 0.15 μm
W = 1.5 μm
L_{ov} = 0.03 μm
$C_{sb0} = C_{db0}$ = 1 fF
V_0 = 0.9 V
V_{SB} = 0.2 V
V_{DS} = 0.8 V

Calculate C_{ox}, C_{ov}, C_{gs}, C_{gd}, C_{sb}, C_{db}

$$\epsilon_{ox} = \underset{SiO_2}{3.9} \times 8.85 \times 10^{-14} \frac{F}{cm} = 3.45 \times 10^{-13} \frac{F}{cm}$$

$$C_{ox} = \frac{\epsilon_{ox}}{t_{ox}} = \frac{3.45 \times 10^{-13} \, F/cm}{3 \cdot 10^{-7} \, cm} = 1.15 \times 10^{-6} \frac{F}{cm^2}$$

$$= 1.15 \times 10^{-14} \frac{F}{\mu m^2}$$

$$= 11.5 \, fF/\mu m^2$$

$$C_{ov} = C_{ox} \cdot W \cdot L_{ov}$$

$$= 11.5 \frac{fF}{\mu m^2} \cdot 1.5 \, \mu m \cdot 0.03 \, \mu m = 0.52 \, fF$$

$$C_{gd} = 0.52 \, fF$$

$$C_{gs} = C_{ov} + \tfrac{2}{3}(W \cdot L \cdot C_{ox})$$

$$= 0.52 \, fF + \tfrac{2}{3}\left(1.5 \, \mu m \cdot 0.15 \, \mu m \cdot 11.5 \frac{fF}{\mu m^2}\right)$$

$$= 2.24 \, fF$$

$$C_{sb} = \frac{C_{sb0}}{\sqrt{1 + \frac{V_{SB}}{V_0}}} = \frac{1 \, fF}{\sqrt{1 + \frac{0.2 \, V}{0.9 \, V}}} = 0.90$$

$$C_{db} = \frac{C_{j0}}{\sqrt{1 + \frac{V_{DB}}{V_0}}} = \frac{1fF}{\sqrt{1 + \frac{1.0}{0.9V}}}$$

$$V_{db} = V_{DS} + V_{SB}$$
$$= 0.8 + 0.2$$
$$= \underline{1.0V}$$

$$\boxed{C_{db} = 0.69\ fF}$$

Exercise 10.2

For the same N channel MOSFET, if I_D = 100 μA and k_n' = 500 μA/V^2. $\left(\frac{W}{L}\right)$

Calculate f_T

$$f_T = \frac{g_m}{2\pi(C_{gd} + C_{gs})} = \frac{g_m}{2\pi(0.52 + 2.24)\,fF}$$

what is g_m? $g_m = \sqrt{2k_n'\left(\frac{W}{L}\right)I_D} = \sqrt{(2)(500\,\frac{\mu A}{V^2})(100\,\mu A)}$

$$g_m = 1\,mA/V$$

$$f_T = \frac{1\,mA/V}{2\pi(2.76\,fF)} = \boxed{57.7\,GHz}$$